

Forder — Book IV

Proposition 20

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

If C, D lie on opposite sides of AB and the angles ACB and ADB are supplementary, then A, C, B, D form a cyclic quadrilateral.

1 Introduction

Forder IV.20 is the converse theorem for the opposite-segment configuration.

In Forder’s source notation, if C, D lie on opposite sides of chord AB and the angles

$$\angle ACB, \quad \angle ADB$$

are supplementary, then A, C, B, D form a cyclic quadrilateral [1].

The production interface uses a relabelled synthetic form. The fixed chord is CD , the points A, B lie on opposite sides of its carrier, and an explicit point X witnesses that

$$\angle XAD$$

is the supplement of

$$\angle CAD.$$

If

$$\angle XAD \cong \angle CBD,$$

then

$$A, C, D, B$$

are concyclic.

The implementation lives in

```
CGJteamLab/Proposition4_20.lean
```

and exposes two public endpoints:

```
hilbert_IV20_circle_converse
hilbert_crossing_rays_transfer
```

These endpoints have different roles. The first is Forder IV.20 itself. The second is the downstream crossing-rays theorem for which the completed circle block was needed in the proportion development.

This distinction is central to the production architecture. The numbered circle theorem is one layer; the crossing-rays transfer built on the completed circle theory is another.

2 Forder's Statement

Forder states:

If C, D are on opposite sides of AB , and $\angle ACB, \angle ADB$ are supplementary, then $ACBD$ is a cyclic quadrilateral.

His printed proof is very short.

Choose a point E on the ray determined by B, C and on the circle through A, B, D . IV.16 says that

$$\angle AEB$$

and

$$\angle ADB$$

are supplementary. By hypothesis,

$$\angle ACB$$

and

$$\angle ADB$$

are also supplementary. Hence

$$\angle ACB \cong \angle AEB.$$

IV.19 then places E on the circle through A, B, C . Since E already lies on the circle through A, B, D , the two circles coincide, and the desired four points are cyclic.

Forder immediately adds the corollary that if

$$AB \perp BC, \quad AD \perp DC,$$

then A, B, C, D are cyclic.

The present production block formalizes IV.20 itself. The printed IV.20.1 corollary is not a separate public endpoint in the current Book IV production sequence.

3 Production Configuration

The production theorem uses the chord CD as the reference line.

Assume:

$$A \mid B$$

with respect to CD .

Let X satisfy

$$X - A - C.$$

Then the ray AD is unchanged and the angle

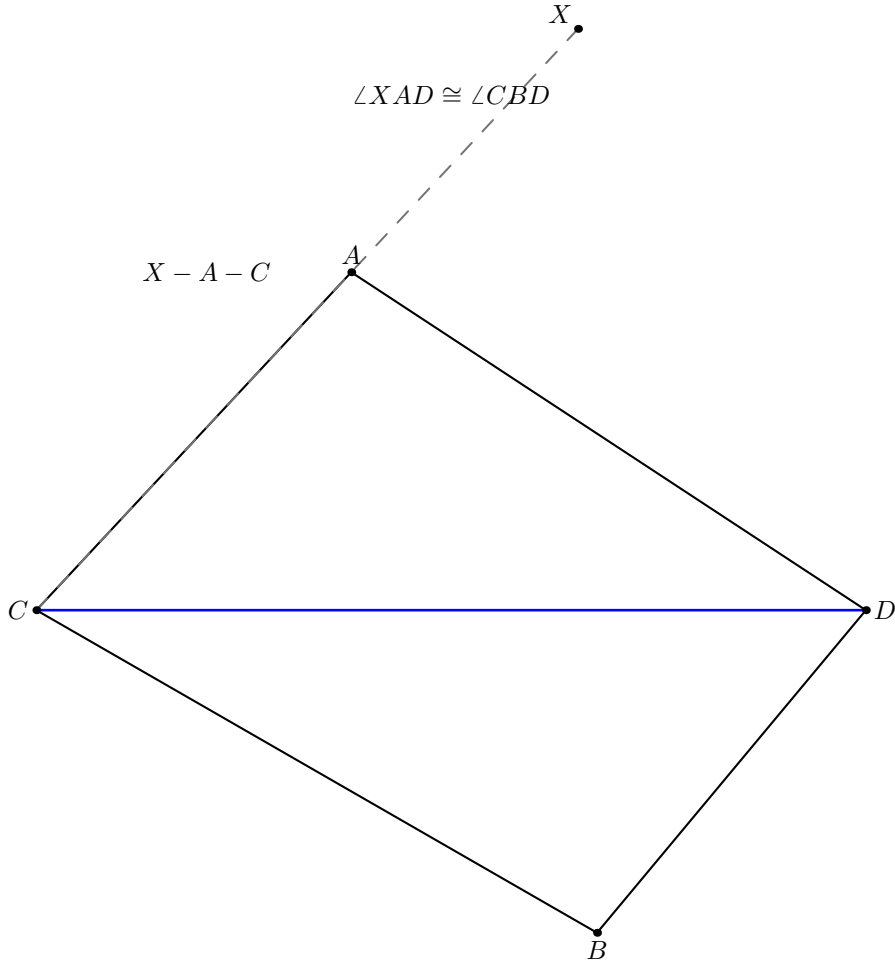
$$\angle XAD$$

is the adjacent supplement of

$$\angle CAD.$$

The hypothesis is

$$\angle XAD \cong \angle CBD.$$



A, B opposite sides of CD

Figure 1: Production form of Forder IV.20. The points A, B lie on opposite sides of chord CD . The point X extends CA through A , so $\angle XAD$ is the supplement of $\angle CAD$. The hypothesis $\angle XAD \cong \angle CBD$ expresses that $\angle CAD$ and $\angle CBD$ are supplementary.

The conclusion is

$$\text{Concyclic4}(A, C, D, B).$$

4 Formal Statement

The interface definition is:

```
def HilbertForderIV20CircleConverse
  [H : HilbertIncidence Geo]
  [_HO : @HilbertOrder Geo H] : Prop :=
  forall C D A B X : Geo.Point,
    forall chord : Geo.Line,
      Ne C D ->
      H.OnLine C chord ->
      H.OnLine D chord ->
      HilbertOppositeSide Geo A B chord ->
      BookZeroSupplement Geo
        X A D
        D C ->
      Geo.AngleCongruent
        X A D
        C B D ->
      HilbertConcyclic4 Geo A C D B
```

The public theorem is:

```
theorem hilbert_IV20_circle_converse
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV20CircleConverse (Geo := Geo)
```

The BookZeroSupplement hypothesis is essential. No numerical equation

$$\angle CAD + \angle CBD = \pi$$

appears in the formal statement. Supplementarity is represented by an actual extension point and a same-ray condition.

5 Source Proof and Production Proof

The source and production proofs establish the same geometric theorem, but their proof organizations are not literally identical.

Forder starts from a circle through A, B, D , chooses a second circle point on the ray through B, C , uses IV.16, then IV.19, and finally circle uniqueness.

The production theorem uses the converse engine already developed for IV.19:

1. construct the circumcircle through C, D, A ;
2. if B lies on it, finish;
3. otherwise analyze the line CB as secant or tangent;
4. eliminate both secant ray orders;

5. eliminate the tangent alternative.

This organization is longer than Forder’s printed proof, but it has a significant formal advantage: the same circle-line and ray-order infrastructure used in IV.19 can be reused without introducing a second circle-intersection existence construction specialized to IV.20.

Thus the formal reconstruction is source-faithful at the level of the synthetic geometry and dependencies, while the proof normalization is adapted to the reusable production API.

6 Constructing the Candidate Circumcircle

Since A, B lie on opposite sides of chord CD , both are off the chord. Therefore C, D, A are noncollinear.

The proof invokes

```
hilbert_triangle_circumcenter_exists_XI
```

to construct a centre K with

$$KC \cong KD, \quad KC \cong KA.$$

This gives a candidate circle containing

$$C, D, A.$$

If B already belongs to the circle, the target concyclicity follows immediately.

Hence all subsequent work takes place under

$$B \notin \text{circle}(K).$$

7 The Carrier CB : Secant or Tangent

Since B is off chord CD ,

$$C \neq B.$$

Let ℓ_{CB} be the line through C, B .

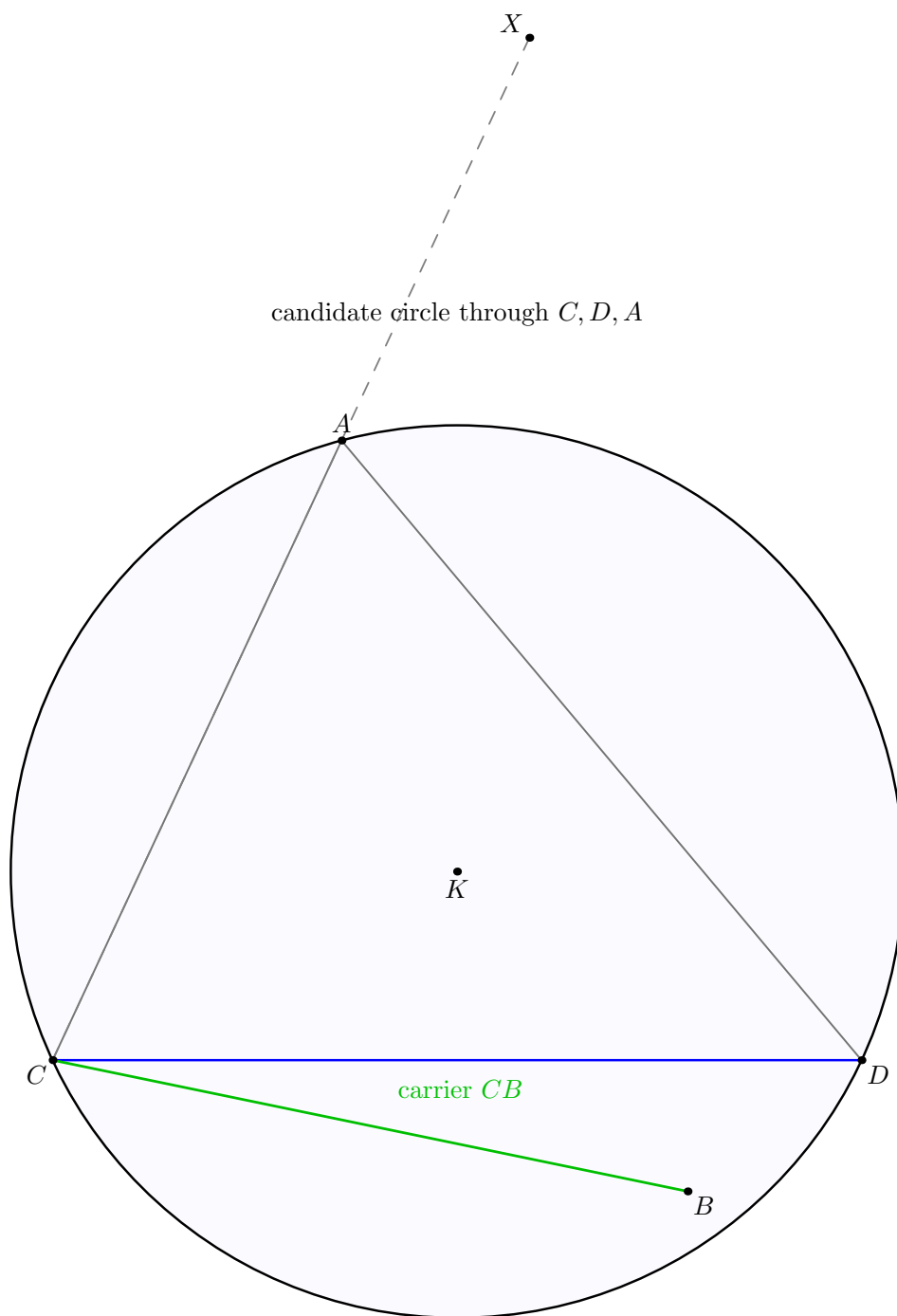
The point D does not lie on ℓ_{CB} , because that would make C, D, B collinear and place B on the chord carrier.

The centre differs from C , so the reusable theorem

```
hilbert_circle_line_second_or_tangent
```

applies.

It returns:



B off circle: secant or tangent

Figure 2: The production converse begins with the circumcircle through C, D, A . The point B is initially outside the logical circle data. If B is not on the candidate circle, the carrier CB is classified as secant or tangent.

$$\begin{cases} \text{a second circle point } E \neq C \text{ on } \ell_{CB}, \\ \text{or} \\ \ell_{CB} \text{ tangent at } C. \end{cases}$$

This is the same structural dichotomy used in IV.19.

8 Secant Branch: Ray-Order Split

Assume there is a second circle point $E \neq C$ on the carrier CB .

Since B is assumed off the circle,

$$E \neq B.$$

The general order theorem

```
hilbert_secant_ray_order_split
```

reduces the line order to:

$$\text{SameRay}(C, B, E)$$

or

$$B - C - E.$$

The two cases use different halves of IV.16.

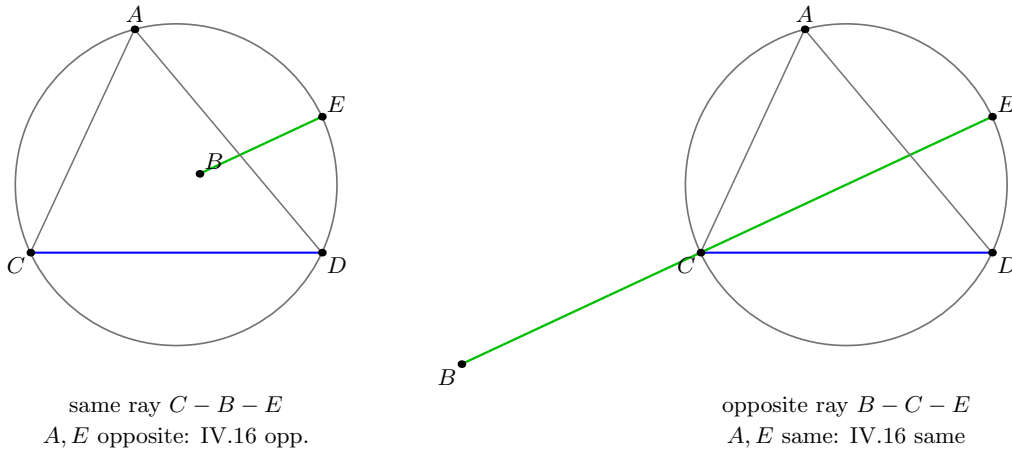


Figure 3: The two secant orders in IV.20. In the same-ray case B, E lie on the same side of chord CD , so A, E are opposite. In the opposite-ray case $B - C - E$, the points B, E are opposite across the chord; since A, B are opposite, A, E are on the same side.

9 Same-Ray Secant Branch

Assume

$$\text{SameRay}(C, B, E).$$

Because B, E lie on the same ray from the chord point C , they lie on the same side of chord CD .

The theorem hypothesis says

$$A \mid B.$$

Therefore

$$A \mid E.$$

Apply IV.16 opposite-side to the circle points

$$C, D, A, E.$$

It produces a point Y such that the angle

$$\angle YED$$

is a supplement of

$$\angle CED$$

and

$$\angle CAD \cong \angle YED.$$

The input at A already says that

$$\angle XAD$$

is a supplement of

$$\angle CAD.$$

After orienting both supplement witnesses consistently, Book Zero theorem 43 transports supplements of congruent angles and yields

$$\angle XAD \cong \angle CED.$$

The IV.20 angle hypothesis is

$$\angle XAD \cong \angle CBD.$$

Hence

$$\angle CED \cong \angle CBD.$$

Now B, E lie on the same ray from C . The reusable theorem

```
hilbert_equal_angles_same_secant_unique
```

forces

$$E = B.$$

But $E \neq B$ because B was assumed off the circle.

Contradiction.

The proposition-specific helper is

```
hilbert_forder_IV20_secant_sameRay_stage7
```

and its essential numbered input is IV.16 opposite-side.

10 Opposite-Ray Secant Branch

Assume

$$B - C - E.$$

The open segment BE meets the chord carrier CD at C , so

$$B \mid E$$

relative to the chord.

But the theorem hypothesis also says

$$A \mid B.$$

Two points opposite to the same point across a line lie on the same side. Hence

$$A \sim E.$$

The general production theorem underlying this parity step is

```
hilbert_two_oppositeSides_sameSide
```

which was promoted from the old stage-labelled development into `HilbertInterfaceIV`.

Apply IV.16 same-side to A, E :

$$\angle CAD \cong \angle CED.$$

The production proof now compares supplements.

The input witness says:

$$\angle XAD$$

is a supplement of

$$\angle CAD.$$

Extend CE through E to a point Y . Then

$$\angle DEY$$

is a supplement of

$$\angle CED.$$

Since

$$\angle CAD \cong \angle CED,$$

Book Zero supplement transport gives

$$\angle XAD \cong \angle DEY.$$

Using the IV.20 hypothesis,

$$\angle XAD \cong \angle CBD,$$

we obtain

$$\angle DEY \cong \angle CBD.$$

Finally $B - C - E$ identifies the ray BC with the ray BE , so

$$\angle CBD = \angle EBD.$$

Therefore

$$\angle DEY \cong \angle EBD.$$

But $\angle DEY$ is an exterior angle of triangle EBD , while $\angle EBD$ is a remote interior angle.

The exterior-angle theorem gives a contradiction.

Thus the opposite-ray secant branch is impossible.

The proposition-specific theorem is

hilbert_forder_IV20_secant_oppositeRay_impossible_stage7

11 Tangent Branch

Assume now that the line CB is tangent to the candidate circle at C .

IV.18 gives the tangent-chord relation:

$$\angle DCB \cong \angle DAC.$$

The IV.20 supplement witness says that

$$\angle XAD$$

is supplementary to

$$\angle DAC.$$

Extend CB through B to a point Z :

$$C - B - Z.$$

Then

$$\angle DBZ$$

is supplementary to

$$\angle CBD.$$

The theorem hypothesis gives

$$\angle XAD \cong \angle CBD.$$

Transporting supplements therefore yields

$$\angle DAC \cong \angle DBZ.$$

Combining with IV.18:

$$\angle DCB \cong \angle DBZ.$$

After reversing the arms at C ,

$$\angle BCD \cong \angle DBZ.$$

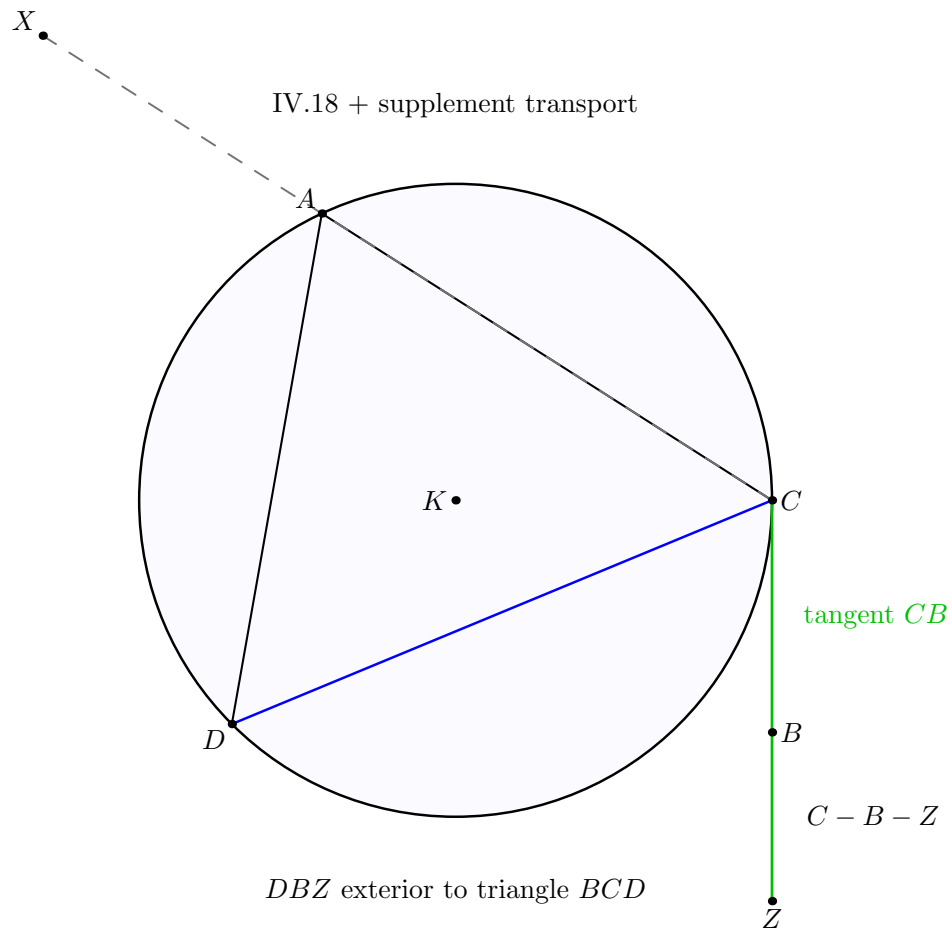


Figure 4: Tangent branch of IV.20. IV.18 identifies the tangent–chord angle at C with $\angle DAC$. Supplement transport then identifies the exterior angle DBZ of triangle BCD with the remote interior angle BCD , which is impossible.

But DBZ is an exterior angle of triangle BCD .

It cannot be congruent to the remote interior angle BCD .

Hence the tangent alternative is impossible.

This is the private helper

```
hilbert_forder_IV20_tangent_impossible_stage7
```

and its essential numbered input is IV.18.

12 Completion of Forder IV.20

The final IV.20 proof is now exhaustive.

Construct the circle through C, D, A .

If B lies on it, finish.

If not, classify the carrier CB :

- secant, same ray: IV.16 opposite-side + supplement transport + same-secant uniqueness force $E = B$;
- secant, opposite ray: IV.16 same-side + supplement transport create an exterior-angle contradiction;
- tangent: IV.18 + supplement transport create an exterior-angle contradiction.

Thus no off-circle position for B is possible.

Therefore

A, C, D, B are concyclic.

The private assembly theorem is

```
hilbert_forder_IV20_circle_converse_stage8
```

and the public wrapper is

```
theorem hilbert_IV20_circle_converse
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV20CircleConverse (Geo := Geo) :=
  hilbert_forder_IV20_circle_converse_stage8
  Geo
  (hilbert_IV16_same_side Geo)
  (hilbert_IV16_opposite_side Geo)
  (hilbert_IV18_tangent_chord Geo)
```

This is an important dependency fact:

IV.20 itself uses IV.16 and IV.18, not IV.19.

The module imports IV.19 for the separate crossing-rays endpoint documented below.

13 IV.20 versus Forder’s Short Proof

The production proof is intentionally more uniform than Forder’s printed argument.

Forder proves IV.20 by choosing a circle point on a specified ray, invoking IV.16, then IV.19, and finally uniqueness of the circle.

The production implementation instead reuses the same “candidate circumcircle + secant/tangent exclusion” architecture as IV.19.

This choice has three formal advantages.

First, all circle-line alternatives are handled by the already verified `hilbert_circle_line_second_or_tangent` theorem.

Second, both secant order cases use the semantic ray-order API already extracted to the interface.

Third, the terminal contradictions are expressed entirely by the established IV.16/IV.18 and exterior-angle machinery, without needing a new theorem about two circles coinciding after sharing three points.

The documentation therefore records both the source proof and the actual production proof rather than silently identifying them.

14 The Crossing-Rays Endpoint

The same production module exports

```
theorem hilbert_crossing_rays_transfer
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertCrossingRaysTransfer (Geo := Geo)
```

The proposition interface is:

```
def HilbertCrossingRaysTransfer
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo] : Prop :=
  forall O A C B D : Geo.Point,
    Not (PrimCollinear Geo A O B) ->
    HilbertSameRay Geo O A C ->
    HilbertSameRay Geo O B D ->
    Geo.AngleCongruent
      O A D
      O B C ->
    Geo.AngleCongruent
      O D C
```

Thus:

$$A, C$$

lie on one ray from O ,

$$B, D$$

lie on a second ray from O , and the crossing configuration is nondegenerate.

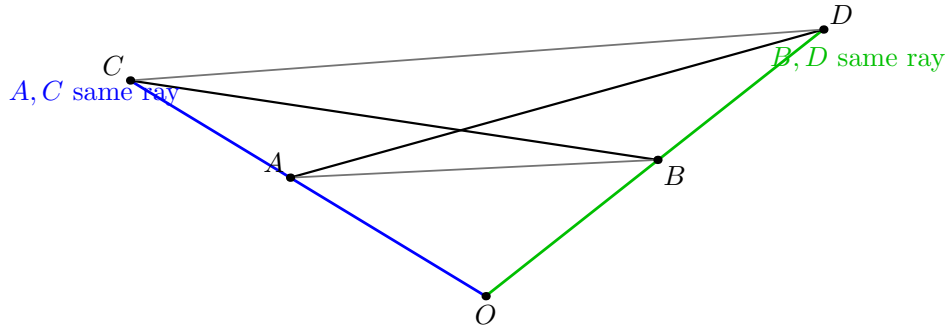
From

$$\angle OAD \cong \angle OBC$$

one proves

$$\angle ODC \cong \angle OAB.$$

$$\text{given: } \angle OAD \cong \angle OBC$$



$$\text{target: } \angle ODC \cong \angle OAB$$

Figure 5: Crossing-rays transfer. The points A, C lie on one ray from O , while B, D lie on the other. One cross-angle congruence, $\angle OAD \cong \angle OBC$, transfers to the complementary congruence $\angle ODC \cong \angle OAB$.

This theorem is not Forder IV.20 itself. It is the downstream result for which the IV.19–IV.20 circle converses provide the concyclicity mechanism.

15 Crossing Chords Become Concyclic

For the proper crossing configuration, the proof first establishes

$$A, C, D, B$$

concyclic.

The local theorem is

hilbert_crossing_chords_concyclic_stage9

It obtains dependency-explicit versions of IV.19 and IV.20 and passes them to the reusable interface theorem

```
hilbert_crossing_chords_concyclic_of_forder
```

The latter already contains the classification of the crossing configuration into:

- same-side chord cases, handled by IV.19;
- opposite-side/supplement cases, handled by IV.20;
- degenerate configurations, handled separately.

This is exactly where IV.19 becomes a genuine dependency of the *module-level endpoint*, even though it is not a dependency of `hilbert_IV20_circle_converse` itself.

16 Degenerate Crossing Cases

The public crossing-rays theorem handles two degenerate possibilities explicitly before entering the circle argument.

16.1 $B = D$

If

$$B = D,$$

the desired conclusion reduces to symmetry of the given angle congruence.

No circle theorem is required.

16.2 $A = C$, with $B \neq D$

If

$$A = C,$$

the circle construction is again unnecessary.

The proof compares triangles

$$OAD \quad \text{and} \quad OBA.$$

Because A, C lie on the same ray and B, D lie on the same ray, the angles at O are congruent.

The second pair is exactly the theorem hypothesis.

The reusable theorem

```
hilbert_AA_third_angle
```

then supplies the required third-angle congruence.

Thus the full crossing theorem does not force a circle into a degenerate configuration where one is not needed.

17 Proper Crossing Case

Assume now:

$$A \neq C, \quad B \neq D.$$

The crossing-chords theorem described above gives

$$\text{Concyclic4}(A, C, D, B).$$

The final angle transfer is then provided by the reusable theorem

```
hilbert_crossing_chords_angle_nondegenerate
```

This theorem converts the concyclic crossing configuration and the given cross-angle congruence into

$$\angle ODC \cong \angle OAB.$$

Hence the proper case proceeds as:

$$\text{IV.19} + \text{IV.20} \implies \text{crossing-chord concyclicity} \implies \text{crossing-angle transfer}.$$

The complete private theorem is

```
hilbert_crossing_rays_transfer_stage11
```

and its closed wrappers finally expose

```
hilbert_crossing_rays_transfer
```

18 Why the Transfer Theorem Lives in Proposition4_20

The proposition `HilbertCrossingRaysTransfer` itself is declared in `HilbertInterfaceIV` because downstream modules must be able to state their contracts without importing the completed circle block.

Its proof, however, cannot live there.

The proper crossing proof needs the completed numbered circle converses: IV.19 and IV.20. Moving the proof into the interface would therefore force the interface to import later numbered propositions and would destroy the acyclic production architecture.

The correct direction is:

$$\text{HilbertInterfaceIV} \longrightarrow \text{Proposition4_19} \longrightarrow \text{Proposition4_20} \longrightarrow \text{hilbert_crossing_rays_transfer}.$$

The proposition interface is early; the theorem proving it is late.

This separation between *statement ownership* and *proof ownership* is intentional.

19 Downstream Role in V.22

The crossing-rays theorem is the circle-geometric kernel needed by the angle-coded proportion calculus.

The downstream production dependency is

Book IV circle theory $\implies$ crossing-rays transfer $\implies$ alternando $\implies$ Hilbert combination rule / Euclid V.22.

Accordingly, the production proportion module imports `Proposition4_20.lean`, because that is the narrow numbered module exporting

```
hilbert_crossing_rays_transfer
```

The proportion theory does not need to import the old monolithic Book IV development.

20 Downstream Role in XI.27

The V.22 result is then consumed by the XI.27 scaling argument.

The dependency route is:

Forder IV.12–IV.20 $\rightarrow$ crossing-rays transfer $\rightarrow$ V.22 $\rightarrow$ XI.27

This is why the Book IV development became production-critical even though its subject matter is planar circle geometry and the eventual application is three-dimensional Euclidean geometry.

The circle block supplies the planar angle-transfer mechanism needed by the proportion layer; the proportion layer then supports the spatial construction.

21 Lean Formalization

The public layer of `Proposition4_20.lean` contains:

```
hilbert_IV20_circle_converse  
hilbert_crossing_rays_transfer
```

The proposition-specific IV.20 layer contains:

```
hilbert_forder_IV20_secant_sameRay_stage7  
hilbert_forder_IV20_secant_oppositeRay_impossible_stage7  
hilbert_forder_IV20_tangent_impossible_stage7  
hilbert_forder_IV20_circle_converse_stage8
```

The crossing endpoint adds private composition theorems:

```

hilbert_crossing_chords_concyclic_stage9
hilbert_crossing_rays_transfer_stage11
hilbert_crossing_rays_transfer_of_IV18_stage55
hilbert_crossing_rays_transfer_complete_stage62

```

The reusable general infrastructure remains in `HilbertInterfaceIV`:

```

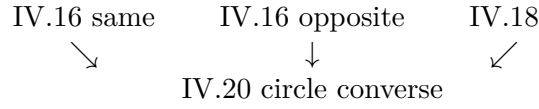
hilbert_circle_line_second_or_tangent
hilbert_equal_angles_same_secant_unique
hilbert_secant_ray_order_split
hilbert_two_oppositeSides_sameSide
hilbert_crossing_chords_concyclic_of_forder
hilbert_crossing_chords_angle_nondegenerate
hilbert_crossing_angle_at_0
hilbert_AA_third_angle

```

The stage-labelled names above are private implementation details. The public API uses semantic production names.

22 Dependency Structure of IV.20 Proper

For the numbered theorem itself:

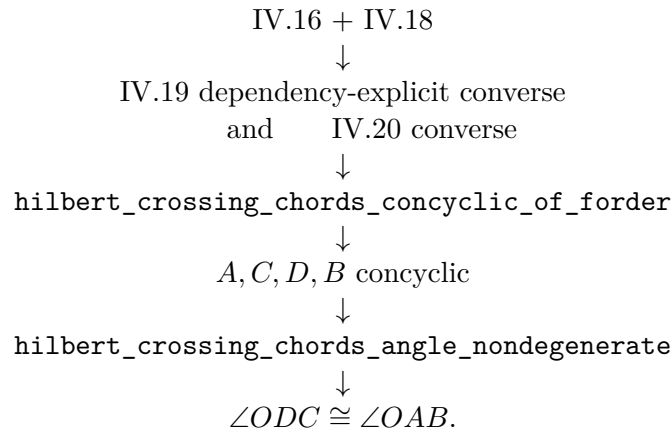


At module-import level, `Proposition4_20.lean` also imports IV.19, but that dependency belongs to the crossing-rays endpoint, not to the numbered IV.20 theorem.

This distinction should be preserved whenever the dependency graph is reported.

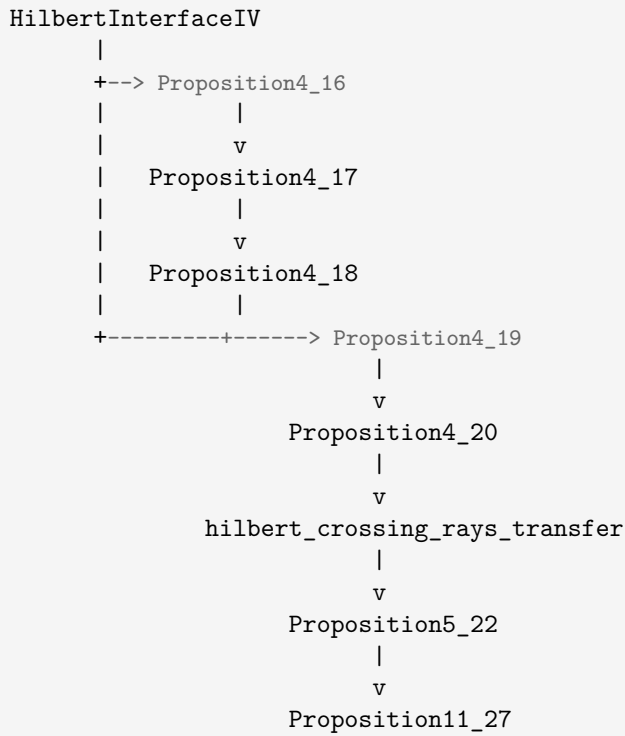
23 Dependency Structure of Crossing-Rays Transfer

The proper crossing branch uses:



Together with the two degenerate cases, this yields the public transfer theorem.

The full module dependency graph at the endpoint is therefore:



The actual direct imports of `Proposition4_20.lean` are:

```

import CGJteamLab.HilbertInterfaceIV
import CGJteamLab.Proposition4_16
import CGJteamLab.Proposition4_18
import CGJteamLab.Proposition4_19

```

24 Foundational Boundary

Neither IV.20 nor crossing-rays transfer introduces a new axiom.

IV.20 uses:

- circumcenter existence for a proper triangle;
- the secant/tangent dichotomy;
- plane separation;
- ray order;
- IV.16;
- IV.18;
- Book Zero supplement transport;

- exterior-angle comparison.

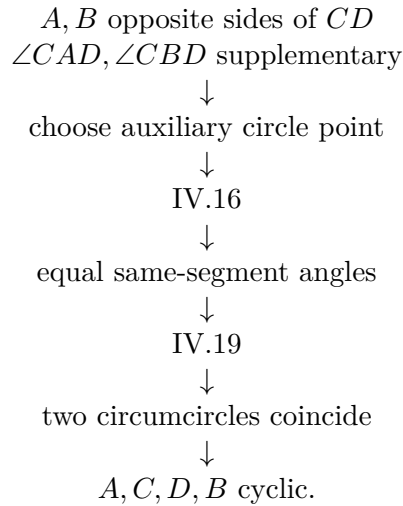
The crossing-rays theorem then uses:

- the completed IV.19 and IV.20 converse machinery;
- the reusable classification of crossing chords;
- AA third-angle transport in a degenerate branch;
- the nondegenerate crossing-chord angle theorem.

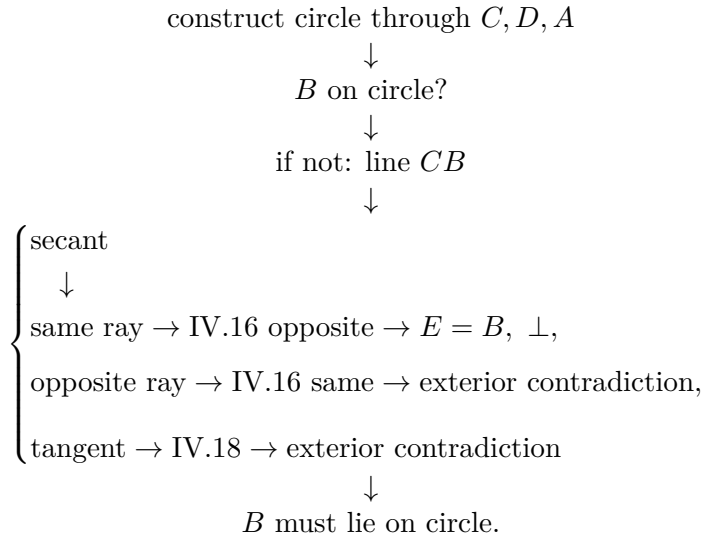
No coordinate slopes, directed-angle arithmetic, determinant concyclicity test, or analytic circle equations are introduced.

25 Classical and Formal Dependency Diagrams

For order IV.20 itself has the source-level shape:



The production proof is:



The crossing endpoint continues:

$$\begin{array}{c}
\text{two ray pairs through } O \\
\downarrow \\
\left\{ \begin{array}{ll} B = D & \text{symmetry,} \\ A = C & \text{AA third angle,} \\ A \neq C, B \neq D & \text{IV.19+IV.20} \rightarrow \text{concylic} \rightarrow \text{angle transfer} \end{array} \right. \\
\downarrow \\
\angle ODC \cong \angle OAB.
\end{array}$$

26 Diagrammatic Audit

Five figures are justified for IV.20 and its endpoint.

1. The synthetic production form of IV.20 with the supplement witness.
2. The candidate circumcircle through C, D, A .
3. A two-panel comparison of the two secant orders.
4. The tangent contradiction using IV.18.
5. The crossing-rays transfer configuration used downstream.

The crossing figure belongs here because the public theorem is exported from the same production module and represents the actual endpoint of the Book IV development.

27 Conclusion

Forder IV.20 closes the pair of circle converses begun by IV.19.

The source proof is short: construct an auxiliary circle point, use IV.16, apply IV.19, and identify the circles.

The production proof adopts the uniform converse engine developed for IV.19. It constructs the circumcircle through three points and eliminates every secant or tangent alternative for the fourth point. The same-ray secant branch uses IV.16 opposite-side; the opposite-ray branch uses IV.16 same-side; the tangent branch uses IV.18.

The module then goes beyond the numbered proposition. Together with IV.19, the completed circle converses prove concyclicity in the proper crossing-rays configuration, and that concyclicity yields the public `hilbert_crossing_rays_transfer` theorem.

This transfer is the actual production bridge from Forder Book IV to the proportion theory:

$$\text{circle geometry} \rightarrow \text{crossing-ray angle transfer} \rightarrow \text{V.22} \rightarrow \text{XI.27}.$$

With IV.20 and the transfer theorem documented, the Forder IV.12–IV.20 production block is mathematically and architecturally complete.

References

- [1] Henry George Forder, *Foundations of Euclidean Geometry*, Cambridge University Press, Cambridge, 1927.